

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 1.6483940962031498, median 1.3187655335638906, std: 1.2762472801143085
Reprojection error (cam1): mean 0.5621328235511502, median 0.4580411885418051, std: 0.4174393928973713
Reprojection error (cam2): mean 0.5671185076621829, median 0.4553134486636886, std: 0.43139437695138294
Gyroscope error (imu0): mean 1.635564791476679, median 1.4222743928476933, std: 1.0077042584758713
Accelerometer error (imu0): mean 0.6879380035434719, median 0.6040660750479905, std: 0.3981822290607212

Residuals

Reprojection error (cam0) [px]: mean 1.6483940962031498, median 1.3187655335638906, std: 1.2762472801143085
Reprojection error (cam1) [px]: mean 0.5621328235511502, median 0.4580411885418051, std: 0.4174393928973713
Reprojection error (cam2) [px]: mean 0.5671185076621829, median 0.4553134486636886, std: 0.43139437695138294
Gyroscope error (imu0) [rad/s]: mean 0.013589695228956832, median 0.011817493034500413, std: 0.008372883682122413
Accelerometer error (imu0) [m/s^2]: mean 0.08671466789921507, median 0.0761426012477282, std: 0.05019091775497139

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[[-0.99974618 -0.01383188 -0.01778377 0.03242388]
 [ 0.01777512 0.00074819 -0.99984173 0.00904589]
 [ 0.01384299 -0.99990406 -0.00050213 -0.08397032]
 [ 0.         0.         0.         1.         ]]
```

T_ic: (cam0 to imu0):

```
[[-0.99974618 0.01777512 0.01384299 0.03341726]
 [-0.01383188 0.00074819 -0.99990406 -0.08352054]
 [-0.01778377 -0.99984173 -0.00050213 0.00957891]
 [ 0.         0.         0.         1.         ]]
```

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)

0.018290170649375784

Transformation (cam1):

T_ci: (imu0 to cam1):

```
[[ -0.99994975 -0.00550521 -0.00837844 0.01801479]
 [ 0.0083938  -0.00277435 -0.99996092 0.00884002]
 [ 0.00548175 -0.999981  0.00282042 -0.08406489]
 [ 0.          0.          0.          1.          ]]
```

T_ic: (cam1 to imu0):

```
[[ -0.99994975 0.0083938 0.00548175 0.0184005 ]
 [ -0.00550521 -0.00277435 -0.999981 -0.08393959]
 [ -0.00837844 -0.99996092 0.00282042 0.00922771]
 [ 0.          0.          0.          1.          ]]
```

timeshift cam1 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)

0.008161466127058516

Transformation (cam2):

T_ci: (imu0 to cam2):

```
[[ -0.99996038 -0.00195472 -0.00868421 -0.03141251]
 [ 0.00868874 -0.00232111 -0.99995956 0.00878467]
 [ 0.00193448 -0.9999954 0.00233801 -0.08432907]
 [ 0.          0.          0.          1.          ]]
```

T_ic: (cam2 to imu0):

```
[[ -0.99996038 0.00868874 0.00193448 -0.03132446]
 [ -0.00195472 -0.00232111 -0.9999954 -0.0843697 ]
 [ -0.00868421 -0.99995956 0.00233801 0.00870868]
 [ 0.          0.          0.          1.          ]]
```

timeshift cam2 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)

0.008159353643298978

Baselines:

Baseline (cam0 to cam1):

[[0.99992108 -0.00940123 -0.00833341 -0.01502125]

[0.00942978 0.99994978 0.00339239 -0.0002263]

[0.0083011 -0.00347071 0.99995952 -0.00033573]

[0. 0. 0. 1.]]

baseline norm: 0.01502670893432391 [m]

Baseline (cam1 to cam2):

[[0.99999365 0.00029582 -0.00355134 -0.04972834]

[-0.00029743 0.99999985 -0.00045161 -0.00008796]

[0.00355121 0.00045266 0.99999359 -0.0003327]

[0. 0. 0. 1.]]

baseline norm: 0.04972952690750182 [m]

Gravity vector in target coords: [m/s^2]

[-0.17786384 -9.80445704 0.09700212]

Calibration configuration

=====

cam0

Camera model: pinhole

Focal length: [887.5915753509161, 885.5818069831571]

Principal point: [641.8664143970775, 362.6550231771033]

Distortion model: radtan

Distortion coefficients: [0.07635563686768929, -0.15066609038705336, 0.0011536539137059101, -0.0031154782024542956]

Type: aprilgrid

Tags:

Rows: 6

Cols: 9
Size: 0.029 [m]
Spacing 0.0087 [m]

cam1

Camera model: pinhole
Focal length: [636.0709551999083, 635.4326819762347]
Principal point: [644.4674854699855, 350.19344257825696]
Distortion model: radtan
Distortion coefficients: [-0.01186432452302042, 0.0034062051255488186, -0.0010128037126135598, 0.0008975748881702977]
Type: aprilgrid
Tags:
Rows: 6
Cols: 9
Size: 0.029 [m]
Spacing 0.0087 [m]

cam2

Camera model: pinhole
Focal length: [635.6262135005884, 634.9023205574423]
Principal point: [647.8529185434957, 350.5218070484684]
Distortion model: radtan
Distortion coefficients: [-0.010030919461651302, -0.000694829470969594, -0.0006969143088979883, 0.0005942923949987655]
Type: aprilgrid
Tags:
Rows: 6
Cols: 9
Size: 0.029 [m]
Spacing 0.0087 [m]

IMU configuration

=====

IMU0:

Model: calibrated

Update rate: 100

Accelerometer:

Noise density: 0.012605012

Noise density (discrete): 0.12605012

Random walk: 0.0018792068913671173

Gyroscope:

Noise density: 0.000830887

Noise density (discrete): 0.00830887

Random walk: 0.0001619178948911119

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

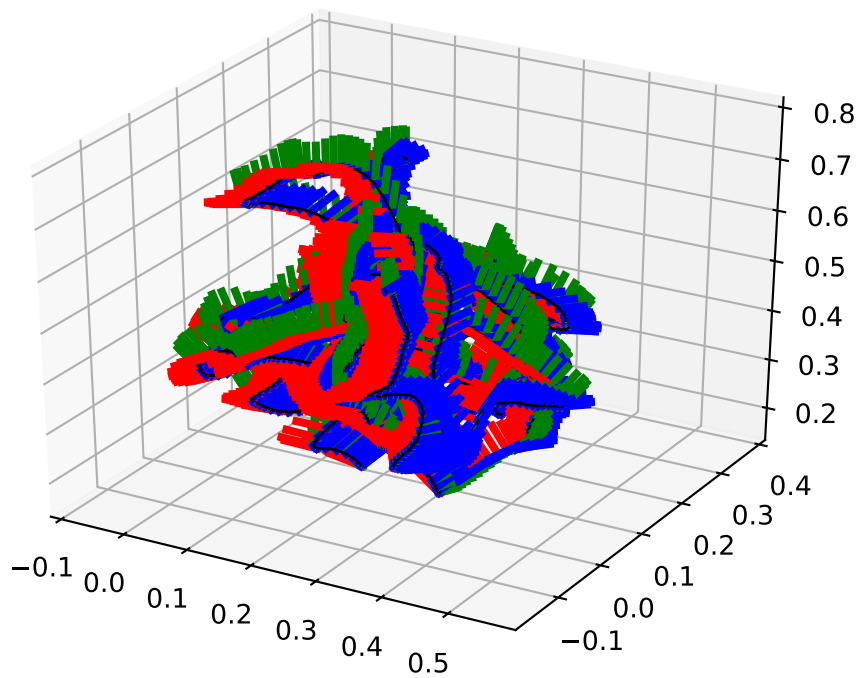
[0. 1. 0. 0.]

[0. 0. 1. 0.]

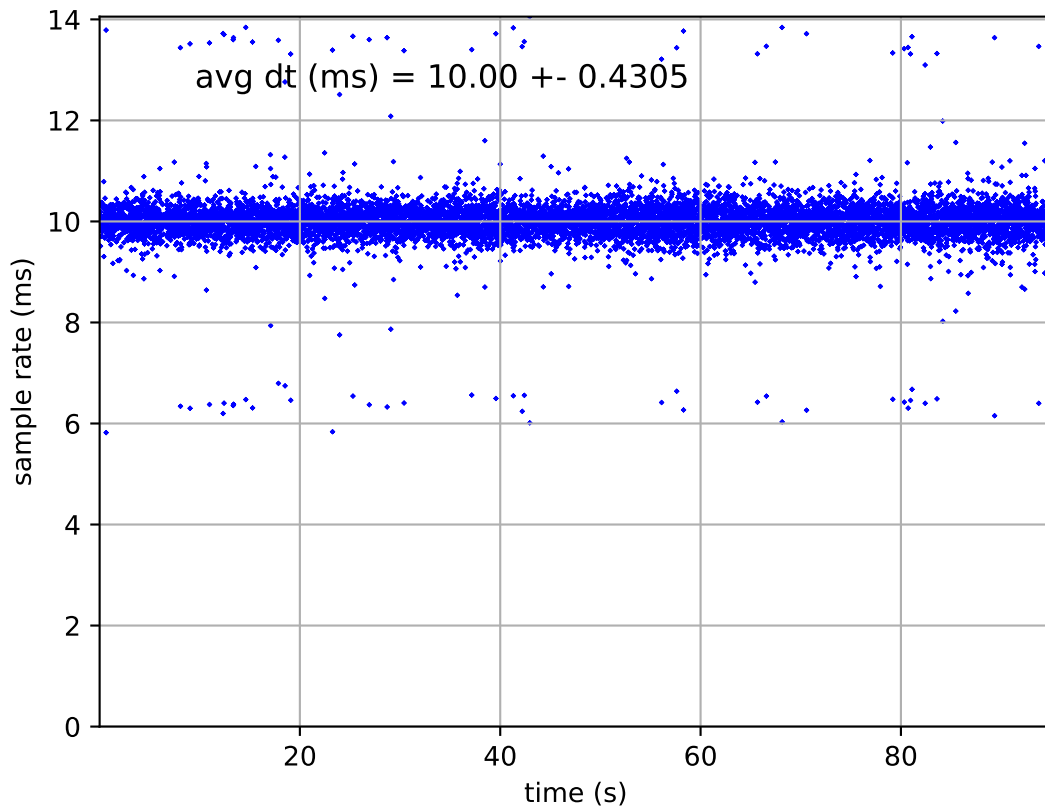
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

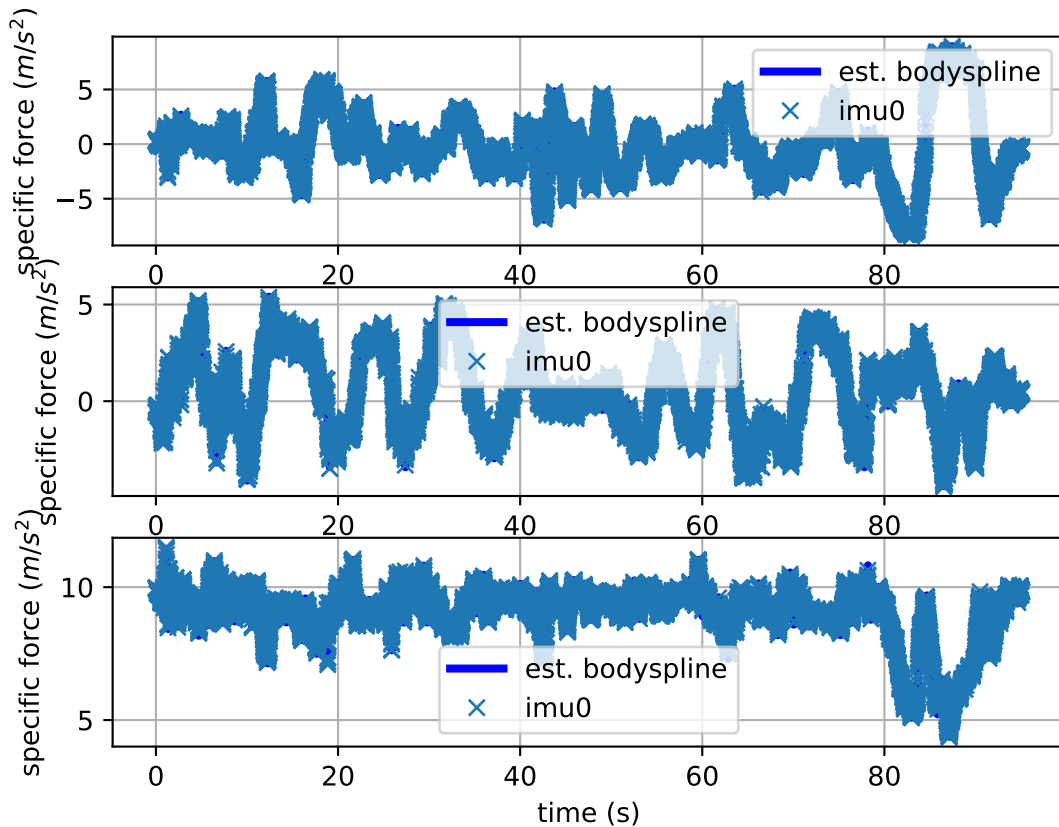
imu0: estimated poses



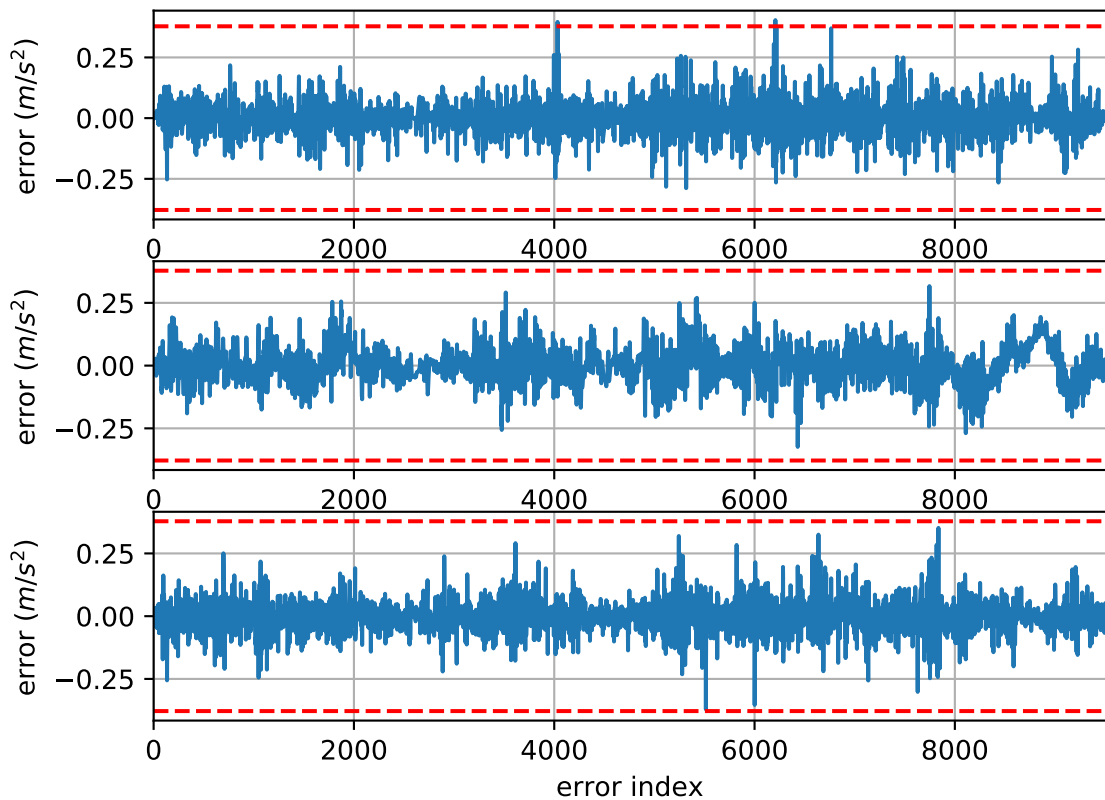
imu0: sample inertial rate



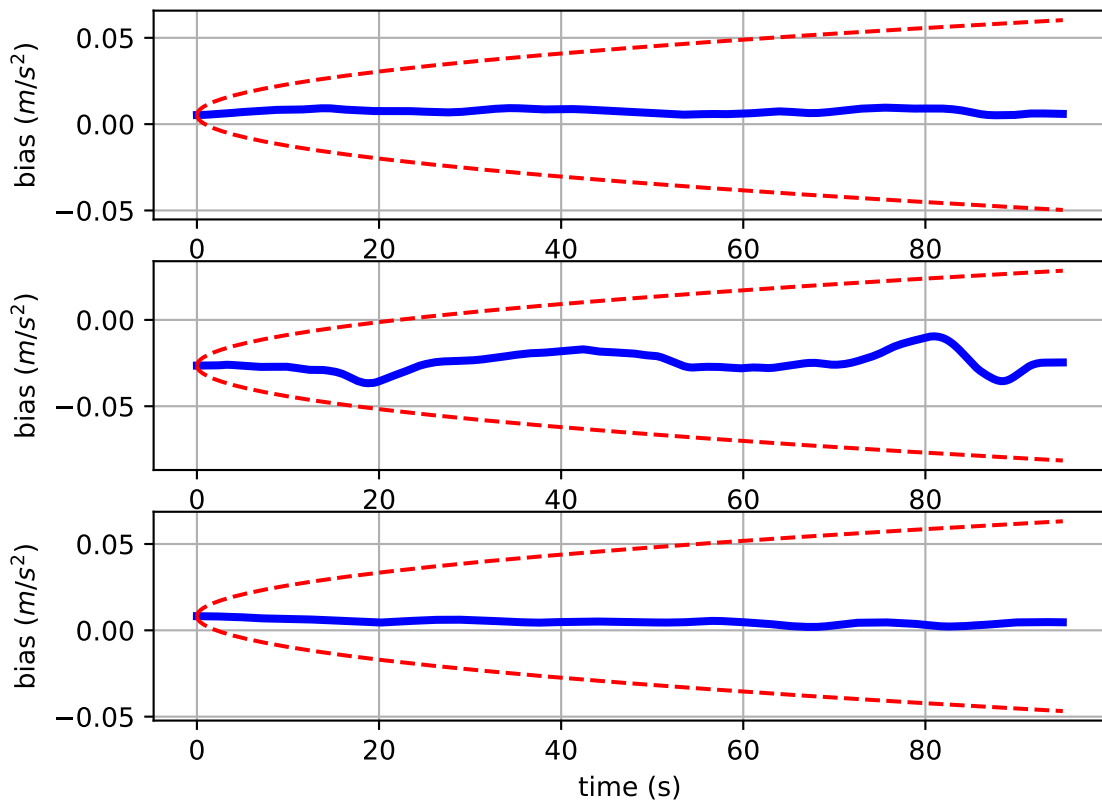
Comparison of predicted and measured specific force (imu0 frame)



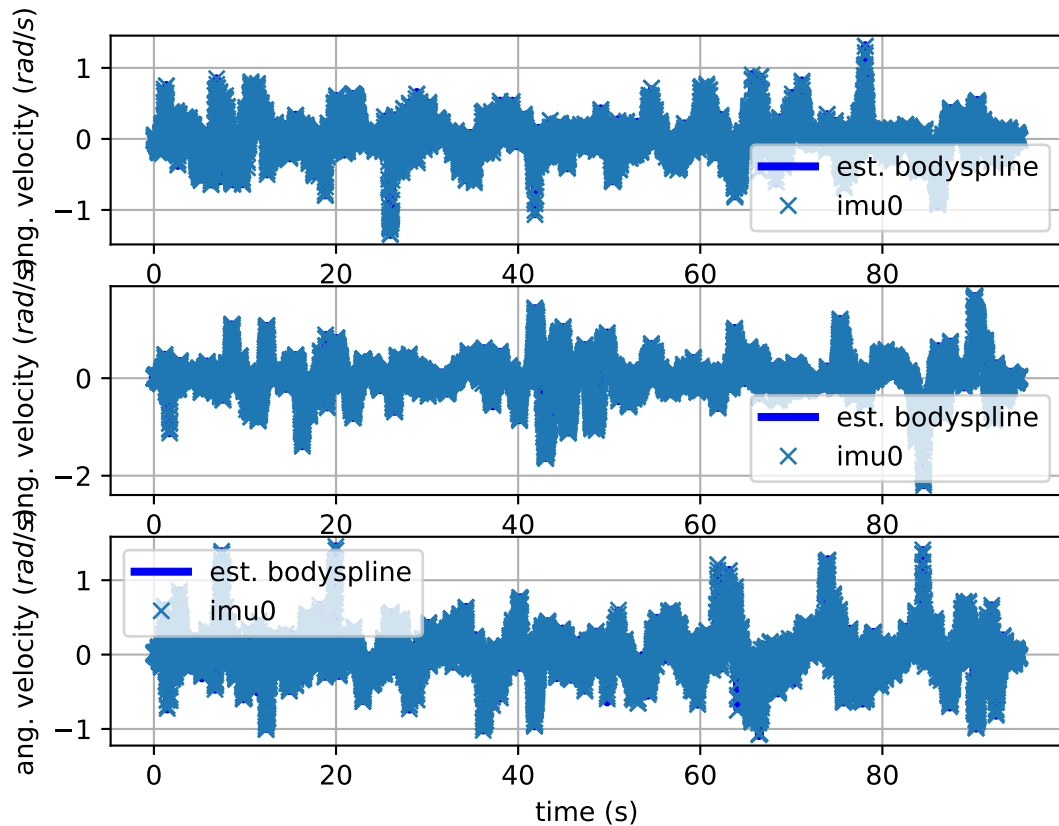
imu0: acceleration error



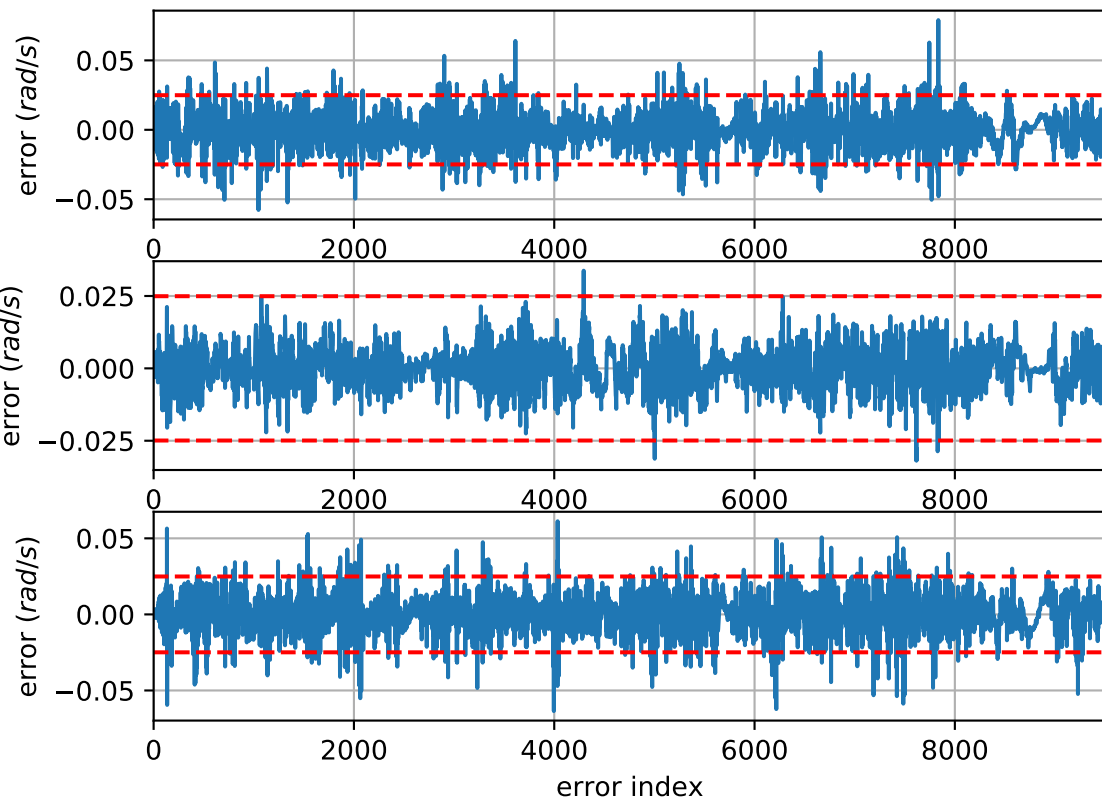
imu0: estimated accelerometer bias (imu frame)



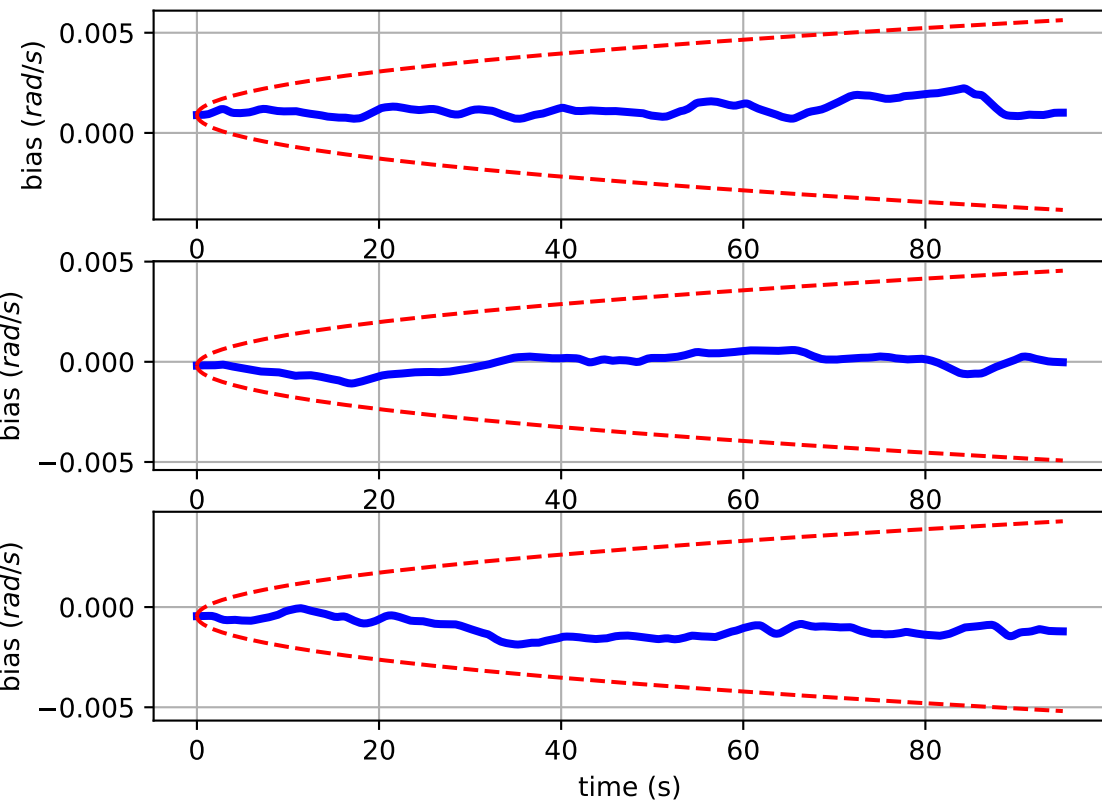
Comparison of predicted and measured angular velocities (body frame)



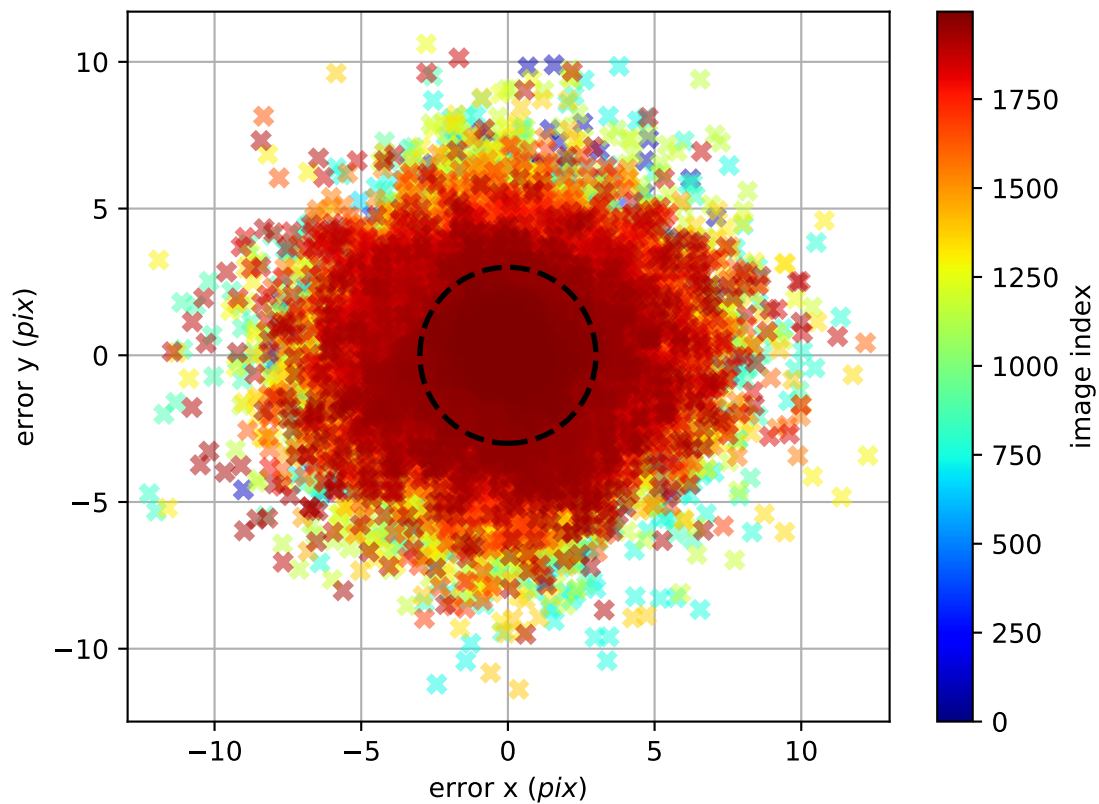
imu0: angular velocities error



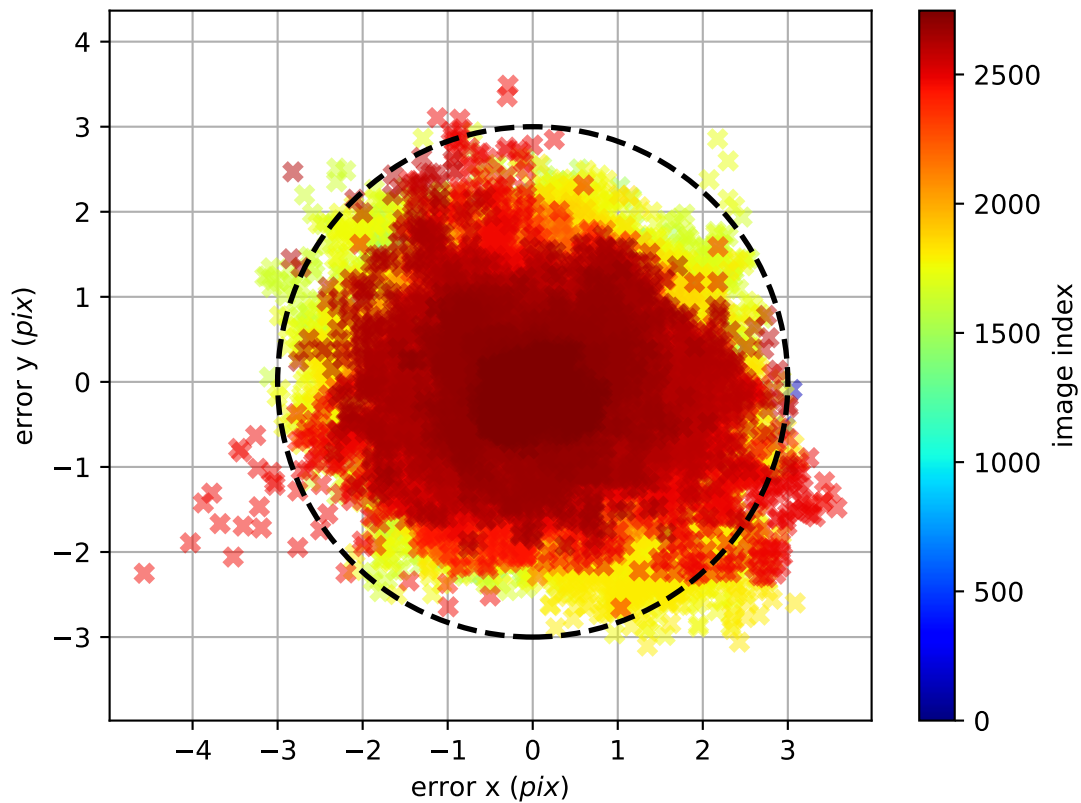
imu0: estimated gyro bias (imu frame)



cam0: reprojection errors



cam1: reprojection errors



cam2: reprojection errors

